

1 Solution — GIAM 1.4

Problem 13

1.1 Problem

Assuming the symbols n, d, q and r have meanings as in the quotient-remainder theorem (GIAM p.29), write expressions for q and r in terms of n and d using floor and/or ceiling notation.

1.2 Recall the Theorem

Theorem 1.4.3 (quotient-remainder). Given integers n and $d > 0$, there are *unique* integers q and r with

$$n = qd + r, \quad 0 \leq r < d.$$

Here $q = n \operatorname{div} d$ is the quotient and $r = n \bmod d$ is the remainder.

1.3 The Expressions

$$\boxed{q = \left\lfloor \frac{n}{d} \right\rfloor, \quad r = n - d \left\lfloor \frac{n}{d} \right\rfloor.}$$

Both use the **floor** function. The quotient is the divided value rounded *down*; the remainder is whatever is left after subtracting off q copies of d .

1.4 Why These Are Right

Set $q = \lfloor n/d \rfloor$ and $r = n - dq$. Two things must be checked: that r is an integer in the range $[0, d)$, and that $n = qd + r$.

- $n = qd + r$ holds by construction: $r = n - dq$ rearranges to $n = qd + r$. And r is an integer, being a difference of integers.
- $0 \leq r < d$. Write the fractional part $\{x\} = x - \lfloor x \rfloor$, which always satisfies $0 \leq \{x\} < 1$. Then

$$r = n - d \left\lfloor \frac{n}{d} \right\rfloor = d \left(\frac{n}{d} - \left\lfloor \frac{n}{d} \right\rfloor \right) = d \cdot \left\{ \frac{n}{d} \right\}.$$

Since $0 \leq \{n/d\} < 1$ and $d > 0$, multiplying through gives $0 \leq r < d$. ✓

So $q = \lfloor n/d \rfloor$ and $r = n - d\lfloor n/d \rfloor$ satisfy *all* the conditions of Theorem 1.4.3, and by the uniqueness clause they are *the* quotient and remainder.

1.5 The Picture

$q = \lfloor n/d \rfloor$ counts the full d -steps below n ; $r = n - qd$ is the leftover. Example: $23 = 4 \cdot 5 + 3$.

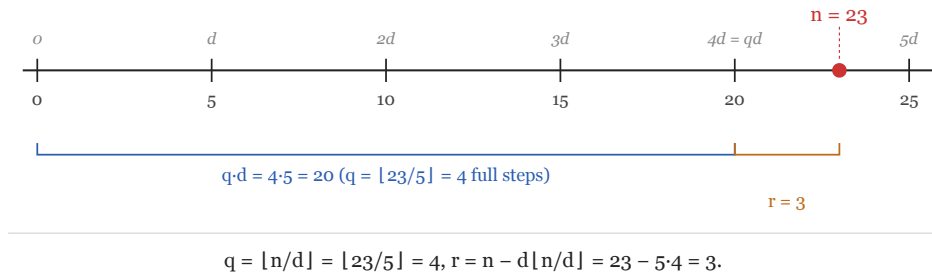


Figure 1.1: On a number line, the multiples $0, d, 2d, \dots$ sit at the ticks. For $n = 23, d = 5$, the largest multiple of 5 not exceeding 23 is $4 \cdot 5 = 20$, so $q = \lfloor 23/5 \rfloor = 4$ full steps, and the leftover is $r = 23 - 20 = 3$. Thus $23 = 4 \cdot 5 + 3$.

The picture is the whole idea: $q = \lfloor n/d \rfloor$ is the *number of complete d -sized steps* you can take from 0 without passing n , and $r = n - qd$ is the *gap* between that last step and n — necessarily smaller than a full step d .

1.6 Remark — Why Floor and Not Ceiling

The remainder must satisfy $0 \leq r < d$, in particular $r \geq 0$. That forces $qd \leq n$, i.e. $q \leq n/d$, so q must be the *largest* integer $\leq n/d$ — which is exactly $\lfloor n/d \rfloor$. Using the ceiling instead, $q = \lceil n/d \rceil \geq n/d$, would give $qd \geq n$ and hence $r = n - qd \leq 0$ — a *non-positive* remainder, violating the theorem's $0 \leq r$ (except when $d \mid n$, where the two agree and $r = 0$). So the sign convention " $r \geq 0$ " is precisely what selects floor over ceiling.

1.7 Remark — These Are Exactly “div” and “mod”

The two expressions are the formal definitions of the operations the textbook names in §1.4.5:

$$n \operatorname{div} d = \left\lfloor \frac{n}{d} \right\rfloor, \quad n \operatorname{mod} d = n - d \left\lfloor \frac{n}{d} \right\rfloor.$$

This is also how `//` (integer division) and `%` (modulo) are *defined* in many programming languages (Python among them) — `n // d` is the floored quotient and `n % d` the matching remainder, chosen so that the identity `n == (n // d) * d + (n % d)` always holds.

1.8 Remark — Floor, Not Truncation, for Negative n

The choice of *floor* (rounding toward $-\infty$) rather than *truncation* (rounding toward 0) is invisible for $n \geq 0$ but essential for negative n — and it is exactly what keeps the remainder non-negative. Take $n = -7$, $d = 3$:

$$q = \left\lfloor \frac{-7}{3} \right\rfloor = \lfloor -2.33\dots \rfloor = -3, \quad r = -7 - 3(-3) = -7 + 9 = 2,$$

giving $-7 = (-3)(3) + 2$ with $0 \leq 2 < 3$. ✓ Had we *truncated* $-7/3$ to -2 (toward zero), we would get $r = -7 - 3(-2) = -1$, a *negative* remainder violating the theorem. So the theorem genuinely requires the floor; truncation-based “integer division” (as in C for negatives) does *not* match the quotient-remainder theorem’s convention.

1.9 Remark — Uniqueness Makes the Formula Unambiguous

The theorem guarantees q and r are *unique*, which is what lets us speak of “the” quotient and “the” remainder — and hence write closed-form expressions for them at all. If, say, $(q, r) = (4, 3)$ and $(q', r') = (3, 8)$ both satisfied $23 = 5q + r$, there would be no single value for $\lfloor n/d \rfloor$ to equal. The constraint $0 \leq r < d$ is what pins the pair down: it forbids “carrying” between q and r (you cannot trade a unit of q for d more in r without pushing $r \geq d$). Floor produces the one admissible pair, and uniqueness certifies it is *the* answer.